NOVA SCHOOL OF SCIENCE & TECHNOLOGY

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CoHabit

Stage n:2



Authors:

80004, Rafaela Ferreira
80063, Miguel Pires
73819, Bernardo Igwe
75153, Bernardo Ascensão

Lab class NºP2

Group Nº 11

Professor:
Teresa Romão

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CoHabit – Stage 2: User and Task Analysis

Problem

People who share a household struggle to coordinate chores, shared expenses, shopping, events, and important information because these are usually spread across chat groups, notes, spreadsheets, and several apps.

CoHabit is an interactive platform that centralizes household management in one place. This stage describes who the users are, what they need to do, and how those needs look in concrete situations, so that the interface designed later fits real use.

Users

CoHabit has three user classes, based on how the household is formed: student roommates, couples, and families. They differ in age, shared responsibilities, and trust between members. The characteristics below come from the Stage 1 target users and our own knowledge.

Class	Characteristics	Main goals	Main pain points
Student roommates	Aged 18-26, high comfort with smartphones, often 3-5 people in a rented flat, tight budgets, irregular schedules, low formal commitment to the group	Split bills and shared purchases fairly, share cleaning duties without arguments, know who is home and who owes what	Conflicts over unequal chores and money owed, messages lost in group chats
Couples	Aged 25-40, comfortable with technology, two members with different routines, shared finances in part, often busy with work/studies	Divide chores evenly, plan shared purchases and events, keep track of joint expenses without it feeling like auditing	Unspoken or unequal workload, forgetting shared appointments, mixing personal and shared spending
Families	Parents (30-65) with children or teenagers, mixed technology skills across ages, routines to coordinate, parents carry most of the organization	Assign age-appropriate chores, coordinate calendars, keep important household information available to everyone	Parents act as the only coordinator, children ignore reminders, information (contacts, passwords, schedules) is scattered or held by one person

Across all classes, the work environment is the home, mostly in short moments: on a phone while standing in the kitchen, in a supermarket, or just before leaving the house. Use is frequent but brief, and members use CoHabit at different times, so one person's actions must be visible to the others.

Tasks

CoHabit should support six high-level tasks. They come from the household activities in the Stage 1 problem statement, plus the set-up task that every household needs first.

Task	Objective	Pre-conditions	Sub-tasks	Exceptions (what can go wrong)	Frequency and constraints
T1. Set up and join a household	Create a shared space and bring all members in	User has the app and an account or invitation	1) Create household; 2) Invite members by link or code; 3) Accept invitation; 4) Set name, members and roles	Invitation link expires or goes to the wrong person; a member leaves or moves out;	Once per household, then rarely (members change); must be quick or people give up
T2. Manage household chores	Share the housework fairly and make sure it gets done	Household exists with at least two members	1) Create chore with frequency; 2) Assign or rotate it; 3) Be reminded; 4) Mark as done; 5) View who did what	Member forgets or refuses the chore; chore is done but not marked; a member is away; disagreement about fairness	Daily to weekly; marking as done must take a few seconds; reminders at suitable times
T3. Track and split shared expenses	Record shared spending and know who owes whom	Household exists; members agreed on shared costs	1) Add expense with amount and payer; 2) Choose who shares it and how; 3) View balances; 4) Register a repayment	Wrong amount or participant entered; one member does not pay; recurring bills (rent, internet) change; disputes about what counts as shared	Several times a week; adding must work at the moment of payment, often in a shop; amounts must be exact

Task	Objective	Pre-conditions	Sub-tasks	Exceptions (what can go wrong)	Frequency and constraints
T4. Manage the shopping list	Buy what the household needs without duplicates or omissions	Household exists	1) Add an item; 2) See and edit the shared list; 3) Check off items while shopping; 4) Share or record what was bought and its cost	Two people buy the same item; list is not updated or not reachable in the shop (weak connection); unclear who is going to buy	Several times a week; used in short moments, often one-handed in a supermarket
T5. Organize household events	Plan events and shared dates so everyone is aware	Household exists	1) Create event with date and place; 2) Invite members; 3) Confirm attendance; 4) See the household calendar	Date clashes; members do not answer; event changes after people confirmed; guests who are not members	Weekly to monthly; must be visible to all; may have a deadline (for example, a birthday or a visit)
T6. Share important household information	Keep key information in one place that every member can find	Household exists	1) Add information (contacts, Wi-Fi, rules, bills dates, notes); 2) Pin or categorize it; 3) Search or open it when needed; 4) Update it	Outdated information; sensitive data (for example, passwords) seen by the wrong person; information buried among many notes	Occasional adding, frequent reading; urgent use (for example, a plumber or an emergency contact) requires quick access

Scenarios

The three scenarios below are concrete and fictional, one per user class. Together they include all six tasks and will be used in the later user evaluation.

Scenario 1: Four students move in together (T1, T4, T3)

Inês, a 20-year-old student, moves into a rented flat in Setúbal with three friends: Tomás, Beatriz and Diogo. On the first evening, she opens CoHabit, creates the household "Flat 3B" and sends an invitation link to the group chat. Tomás and Beatriz accept in a minute, but Diogo's link does not open on his phone, so Inês generates an invitation code and sends it again. The next day Beatriz is at the supermarket with a basket in one hand. She opens the shared shopping list, sees that Tomás has already added washing-up liquid and rice, and checks the items off as she puts them in the basket. At checkout, she pays € 32.40, adds it as a shared expense split equally among the four members, and the app shows that each of the others owes her € 8.10.

Scenario 2: A couple divides chores and plans a dinner (T2, T3, T5)

Rui and Ana, both 29 and working full time, live together and feel that Ana does most of the cleaning. On Sunday evening, Rui opens CoHabit and creates five weekly chores: bathroom, kitchen floor, laundry, bins and groceries. They agree to alternate the chores and the app sets a rotation. During the week, Ana marks the laundry as done, but Rui forgets the bins and receives a reminder the next evening. On Friday, they decide to host four friends on Saturday. Rui creates an event with the date and time, invites Ana, and she confirms on her phone. Ana then pays 58 euros for food and registers it as a shared expense so their balance stays up to date.

Scenario 3: A family organizes a birthday and finds key information (T2, T5, T6)

Sofia, a 45-year-old mother, lives with her husband Carlos and their teenagers Mariana (16) and João (13). On Monday, she creates age-appropriate chores in CoHabit: João takes out the recycling, and Mariana sets the table on weekdays. João gets a notification and marks his chore as done once it is finished. Sofia then creates an event for Carlos's birthday on Saturday and invites all members, who confirm in the app. During the week, a pipe starts leaking. Carlos opens the household information section, finds the plumber's number that Sofia saved there earlier, and calls without asking anyone. Sofia also updates the Wi-Fi password after changing the router, so that Mariana and João can reconnect their devices.

Interviews

We will interview at least three people, one per user class, and summarize what they said.

Teodora, 23, Erasmus student living with a roommate

The interview with Teodora reveals that the main difficulties in her shared living are *not necessarily major conflicts, but rather the absence of organization and structure* in day-to-day household management. According to her responses, **household responsibilities** are not formally assigned and cleaning tasks are generally performed whenever someone has time or notices that something needs to be done. While this informal approach may work to some extent, it also creates situations where one roommate may feel that the distribution of effort is not entirely fair. This perception is confirmed when she mentions that she has sometimes felt that cleaning time was not allocated equally between the two residents.

The **management of shared expenses** also emerged as a significant pain point. Although rent and utilities are split equally, confusion sometimes arises when paying for groceries or household supplies that are shared by both residents. The interviewee reported situations in which it became unclear who paid for what, or who owed money to whom after purchasing cleaning products or cooking together. This suggests that expense-tracking features could provide substantial value. In this context, *the need is not for complex financial management, but rather for a lightweight way to keep track of shared purchases*, debts, and reimbursements without requiring users to manually maintain records through messaging applications.

Regarding **communication**, the responses indicate that WhatsApp and face-to-face conversations *are already sufficient for most interactions.* However, important information still needs to be exchanged, particularly messages from the landlord or matters related to apartment maintenance. This suggests that the interviewee does not necessarily need another communication platform, but could benefit from having important household information centralized in a dedicated place.

The **polling feature** received a more moderate reaction. The interviewee acknowledged that polls could be useful, particularly in households with more residents, and mentioned examples such as deciding on cleaning priorities or finding a suitable day for everyone to clean together. However, unlike the responses regarding chores and expense tracking, her enthusiasm was noticeably lower.

Conclusions: the interview highlights an important design insight: users are not necessarily looking for more ways to communicate, since existing messaging applications already fulfil that role. Instead, they need tools that introduce organization, transparency, and accountability into shared living spaces. Based on these findings, the application's core value proposition should focus on simplifying household coordination through shared chores, automatic reminders, expense management, and scheduling tools. Features such as polls and announcements can still be included, but should likely be treated as complementary functionalities rather than the primary focus of the product.

Maria, 61, linguistics teacher and single-mother living with her two adult sons

The interview with Maria, a working mother living with her two adult sons, presents a somewhat different perspective on cohabitation from the student interview.

Household responsibilities are handled in a largely informal manner. Maria explains that chores are not assigned and are typically completed by whoever is home and has time, although she acknowledges that she performs most of them herself. Interestingly, she does not characterize this arrangement as unfair despite admitting that the workload is not equally distributed. Instead, she frames it as part of her role as a parent, while also expressing a desire for more help from her sons. This indicates that features such as **chore assignment and automatic chore rotation** may have limited appeal in family settings where household roles are already well established. However, reminder notifications received a clearly positive response, suggesting that *prompting household members to contribute may be more valuable than formally distributing tasks*.

The interview also reveals that **expense tracking**, one of the strongest needs identified in the previous interview, is considerably less relevant here. Maria manages most household finances herself, and occasional contributions from her oldest son are handled informally. Because the *family does not actively track debts* or reimbursements between members, there have been no significant misunderstandings regarding shared expenses. This suggests that bill-splitting features may be much more useful for groups of unrelated adults living together than for families sharing a household budget.

Another interesting finding concerns **decision-making**. Maria reports frequent difficulties reaching decisions on matters such as household purchases, lifestyle, and family finances. Despite this, she expressed *little interest in a polling or voting system*. This may indicate that the problem is not a lack of a mechanism for voting, but rather the complexity of family discussions themselves. In a family environment, conversations and negotiations may be preferred over formal voting processes, even when reaching agreements is challenging.

Conclusions: The findings also highlight the importance of user adoption. Even if a feature is technically useful, it may provide little value if some household members are unwilling to use the application consistently. For this reason, we once again conclude that the app should prioritize simplicity and low effort, ensuring that users can benefit from it without significantly changing their existing habits. Overall, this interview supports features aimed at coordination and reminders, while raising questions about the usefulness of more structured functionality in family-oriented households.